CLASS 9 — MATHS · FORMULA SHEET

Chapter 6: **Perimeter and Area** | Quick revision, one page

Circle and arc ($\pi = \frac{22}{7}$ or 3.14)

- Circumference $= 2\pi r = \pi d$; area $= \pi r^2$.
- Arc of a circle with angle θ at the centre: length $= \dfrac{\theta}{360} \times 2\pi r$.
- Area of a ring (path of width w round a circle of radius r): $\pi\left[(r+w)^2 - r^2\right]$.

Parallelogram and triangle

- Parallelogram: area $=$ base $\times$ height; perimeter $= 2(a+b)$.
- Triangle: area $= \frac{1}{2} \times$ base $\times$ height.
- Equilateral triangle of side a: area $= \dfrac{\sqrt{3}}{4}a^2$.
- Altitude on a side $= \dfrac{2 \times \text{area}}{\text{that side}}$.

Heron's formula

- $s = \dfrac{a+b+c}{2}$.
- Area $= \sqrt{s(s-a)(s-b)(s-c)}$.
- Example: sides $13, 14, 15$ give $s = 21$ and area 84.

Brahmagupta's formula (cyclic quadrilateral)

- $s = \dfrac{a+b+c+d}{2}$.
- Area $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$; it applies only when the quadrilateral is cyclic.
- Units: lengths in cm or m, area in cm^2 or m^2; cost $=$ area $\times$ rate per unit area.

MASTERMIND